

Module 03 : Identity and Access Management

Root Account (Root User)

The "root user" refers to the account created when signing into AWS using the email and password associated with the account. This user has unrestricted access to the AWS account and can perform critical actions such as:

- Changing the support plan
- Modifying payment methods
- Closing the AWS account
- Transferring ownership of the account

The root user functions is similar to an admin on a computer, with the ability to install, modify, or delete any aspect of the system.

Given the level of access, it is crucial to secure the root user. This can be done by enabling multi-factor authentication (MFA) or two-factor authentication (2FA) to add an extra layer of security.

Multi Factor Authentication for Users

Enabling MFA for AWS Users:

1. After logging into your AWS account, search for the IAM service.
2. Under **Security Recommendations**, choose to add MFA for the root user.
3. Alternatively, in the top-right corner of the dashboard, click on **Security Credentials**.
4. Under **Multi-factor authentication (MFA)**, select **Assign MFA device**.

Types of MFA Devices:

1. **Authenticator App**: Applications like Google Authenticator, Microsoft Authenticator, etc., generate one-time codes.

2. **Security Key:** A physical device (similar to a USB drive) that must be connected to your computer or laptop.
3. **TOTP (Time-based One-Time Password Token):** A hardware device (resembling a USB drive) that continuously displays a six-digit code, which must be entered during login. This functions like an RSA token.

Note: The **Security Key** is considered the most secure option, as it physically needs to be present. In contrast, apps or TOTP devices can expose their codes if left unattended or forgotten in places like home.

IAM User: Identity and Access Management.

IAM (Identity and Access Management) allows us to create users and groups, and manage their permissions to access AWS resources.

Example:

In a large project, multiple resources may be involved, and IAM helps us assign specific permissions based on roles:

1. AWS Engineer:

- For handling **EC2/server issues**, we create a user with their own username and password, but restrict their access to only specific resources. For example, they can be given **EC2 full access**, but any attempt to perform actions outside of this scope would be denied.

2. Database Administrator:

- For resolving **database issues**, a separate user with individual credentials is created, and they are granted **RDS full access**.

3. Monitoring:

- To **monitor AWS resources**, a user is created with **read-only access**, allowing them to view resources but not make any changes.

Key Concepts:

- **Least Privilege Mechanism:** This principle ensures that users are granted the minimum permissions necessary to perform their job functions. IAM allows us to enforce this by

restricting user access to specific operations and resources based on their roles.

💡 **Important for Certification:** Understanding and implementing least privilege is crucial for security best practices.

💡 **Note:** In real-world organizations, users rarely operate as root users. Instead, they work as IAM users with specific permissions tailored to their role.



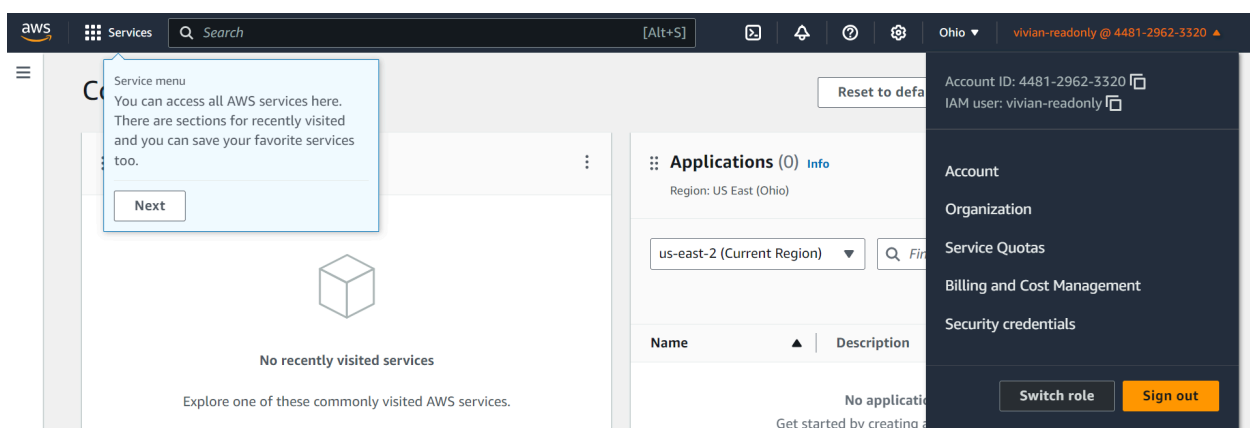
An AWS account can only have one root user; multiple root users are NOT allowed for a single account. It's not recommended to share root credentials with everyone. By creating an IAM multiple users and manage their access. This is the main advantage of IAM.

IAM is completely FREE always; no limitation.

How to Identify if You Are Logged in as a Root User or IAM User

- After logging in, click on the account information in the top-right corner of the console.
- If only the 12-digit account ID is displayed, you are logged in as the **root user**.
- If the display shows "username@account-id," you are logged in as an **IAM user**.

This distinction helps you quickly identify the type of account access you are using.



Creating read-only IAM user

1. Log in to the AWS Management Console:

- Sign in using your root account or an IAM account with administrative privileges.
2. **Navigate to the IAM Dashboard:**
 - From the AWS Management Console, type "IAM" in the search bar and select **IAM**
 3. **Create a New User:**
 - In the IAM dashboard, click on **Users** in the left-hand menu.
 - Click the **Create User** button.
 4. **Set User Details:**
 - Enter a **Username** for the new user.



When creating users in AWS, there are two primary types of access:

1. **Management Console Access** – Allows the user to log in via a web browser.
2. **Programmatic Access** – Enables access through the AWS CLI, SDKs, or APIs (covered later in the course).

For now, we're focusing on **Management Console Access**.

Next, You will be prompted with: **“Are you providing console access to a person?”**

You will see two options:

- **Specify a user in IAM Identity Center** – Identity center topic is covered later in the course.
- **I want to create an IAM user** – Choose this option for now and proceed to set credentials for the new IAM user.

4.1 Enable Console Access

- Check the box: **“Provide user access to the AWS Management Console – optional”** to allow web-based login.
- Select the user type: **“I want to create an IAM user”** – Choose this option for now and proceed to set credentials for the new IAM user.

- **Console Password:** Set the user's console password, either by choosing an auto-generated password or specifying your own password.

Search [Alt+S] Global prod

ers > Create user

Step 3
○ Review and create
Step 4
○ Retrieve password

User name

iam-admin-billing-test

The user name can have up to 64 characters. Valid characters: A-Z, a-z, 0-9, and + = , . @ _ - (hyphen)

☒ **Provide user access to the AWS Management Console - optional**
If you're providing console access to a person, it's a [best practice](#) to manage their access in IAM Identity Center.

Are you providing console access to a person?

User type

☐ **Specify a user in Identity Center - Recommended**
We recommend that you use Identity Center to provide console access to a person. With Identity Center, you can centrally manage user access to their AWS accounts and cloud applications.

☒ **I want to create an IAM user**
We recommend that you create IAM users only if you need to enable programmatic access through access keys, service-specific credentials for AWS CodeCommit or Amazon Keyspaces, or a backup credential for emergency account access.

Console password

☒ **Autogenerated password**
You can view the password after you create the user.

☐ **Custom password**
Enter a custom password for the user.

- Must be at least 8 characters long
- Must include at least one uppercase letter (A-Z)
- Must include at least one lowercase letter (a-z)
- Must include at least one number (0-9)
- Must include at least one non-alphanumeric character (! @ # \$ % ^ & * () _ + = { } |)

☐ Show password

☒ **Users must create a new password at next sign-in - Recommended**

If you are creating programmatic access through access keys or service-specific credentials for AWS CodeCommit or Amazon Keyspaces, you can generate them after you create this IAM user.
[Learn more](#)

Cancel Next

5. Set Permissions:

- On the **Permissions** page, choose **Attach policies directly**.
- Search for and select the policy named "**ReadOnlyAccess**". This grants the user read-only access to all AWS services.
- Skip this as shown in the lecture video - "**Set permissions boundary - optional**"



When assigning permissions to a new IAM user, you have 3 options:

1. Add User to a Group

Create a new group (or use an existing one), assign policies to that group, and then add the user to it.



Recommended for managing permissions at scale.

2. Copy Permissions from an Existing User

Duplicate the permission set from another IAM user.



Useful for quickly onboarding users with similar roles.

3. Attach Policies Directly

Assign one or more IAM policies directly to the user.



Not ideal for long-term management, especially in larger environments.

6. Review and Create:

- Review the user's details on the **Review** page to ensure everything is correct.
- Click **Create User** to finalize the process.

7. Provide Access Information:

- After creating the user, AWS will display access credentials (if console access was enabled) and login details for console access.
- Download or copy this information and share it with the user securely.

The new IAM user now has read-only access across your AWS account.

Creating a storage administrator user group with full access to S3 only

1. Log in to the AWS Management Console

- Sign in with your root or an IAM user with administrative privileges.

2. Navigate to the IAM Dashboard

- In the AWS Management Console, type **IAM** in the search bar and select **IAM**.

3. Create a New Group for S3 Administrators

- In the IAM dashboard, click on **User Groups** in the left-hand menu.
- Click **Create Group**.

Steps for Creating the Group:

- **Group Name:** Enter a name like `S3-Admins`.
- **Attach Policies:**
 - In the policy search bar, type `AmazonS3FullAccess`.
 - Select the policy **AmazonS3FullAccess** to give the group full access to all S3 buckets and objects.
- Click **Create Group**.

4. Create the IAM User (s3Admin-user)

- On the IAM dashboard, click **Users** in the left-hand menu.
- Click **Add User**.

Steps for Adding the User:

- **User Name:** Enter the username `s3Admin-user`.
- **Access Type:**
 - If console access is needed, select **AWS Management Console access** and choose either an autogenerated or custom password.
 - If programmatic access (API, CLI, or SDK access) is required, select **Programmatic access**.
- Click **Next: Permissions**.

5. Assign the User to the S3-Admins Group

- On the **Permissions** page, choose **Add user to group**.
- Select the group you just created, **S3-Admins**.
- Click **Next**.

6. Review and Create the User

- Review the user details to ensure everything is correct.
- Click **Create User** to finish.

7. Provide Access Information

- If you enabled programmatic access, download the **access keys** (Access Key ID and Secret Access Key).
- Provide the user with their console login details and/or access keys.

Creating IAM user with administrative access

The **root user** has complete, unrestricted access to all resources and settings in an AWS account. In contrast, an **IAM user with administrative access** has nearly the same level of control but with certain limitations. While the IAM administrator can manage services and resources across AWS, they cannot perform sensitive account management tasks such as:

- Closing the account
- Changing the support plan
- Modifying the payment methods

These specific actions are reserved for the root user only.

1. Log in to the AWS Management Console:

- Sign in using your root account or an IAM account with administrative privileges.

2. Navigate to the IAM Dashboard:

- From the AWS Management Console, type "IAM" in the search bar and select **IAM**

3. Create a New User:

- In the IAM dashboard, click on **Users** in the left-hand menu.
- Click the **Create User** button.

4. Set User Details:

- Enter a **Username** for the new user.
- Check the box “Provide user access to the AWS Management Console - *optional*” for **AWS Management Console access** (for console access).
- If console access is selected, set the user's console password, either by choosing an auto-generated password or specifying your own password.

5. Steps for Adding the User:

- **User Name:** Enter the desired username (e.g., `admin-user`).
- **Access Type:**
 - For management console access, check **AWS Management Console access** and choose either an autogenerated or custom password.
 - For programmatic access, check **Programmatic access** to allow API, CLI, or SDK access.
- Click **Next: Permissions**.

6. Assign Administrator Access

- On the **Permissions** page, choose **Attach policies directly**.
- In the policy search bar, type `AdministratorAccess`.
- Select the policy **AdministratorAccess**, which grants full permissions across all AWS services.
- Click **Next**.

7. Review and Create the User

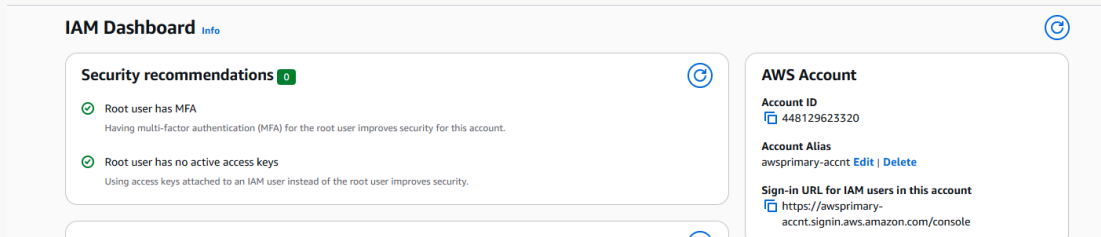
- On the **Review** page, verify that the correct username and policies are assigned.
- Click **Create User** to complete the process.

8. Provide Access Information

- If you enabled programmatic access, download the **Access Key ID** and **Secret Access Key**.
- For console access, note the login URL and credentials for the user.



- All IAM users share a common login URL. For example:
<https://xxxxxxxxxxxxxxxxx.signin.aws.amazon.com/console>



- It's important to note that **renaming an IAM username is NOT supported**.

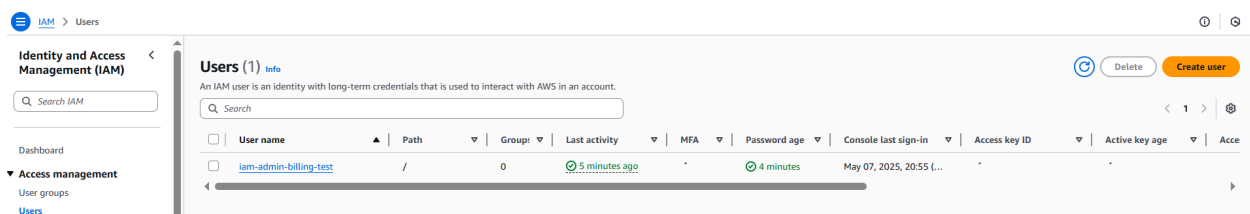
Assignments

Task 1: Create an IAM user, Add "AdministratorAccess" policy. Login and try to access "Billing information". It will deny.

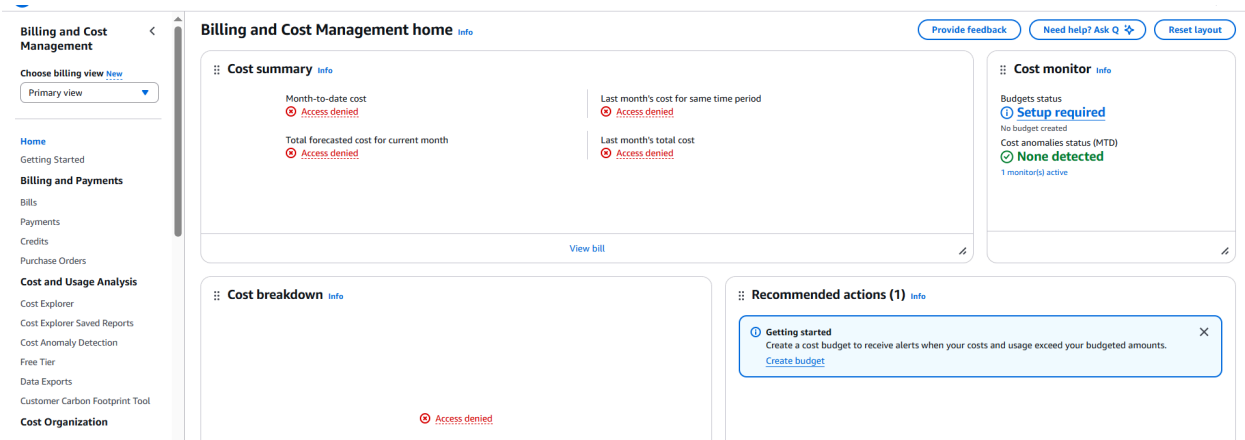
--> provide Billing access to the above IAM user.

IAM User Creation:

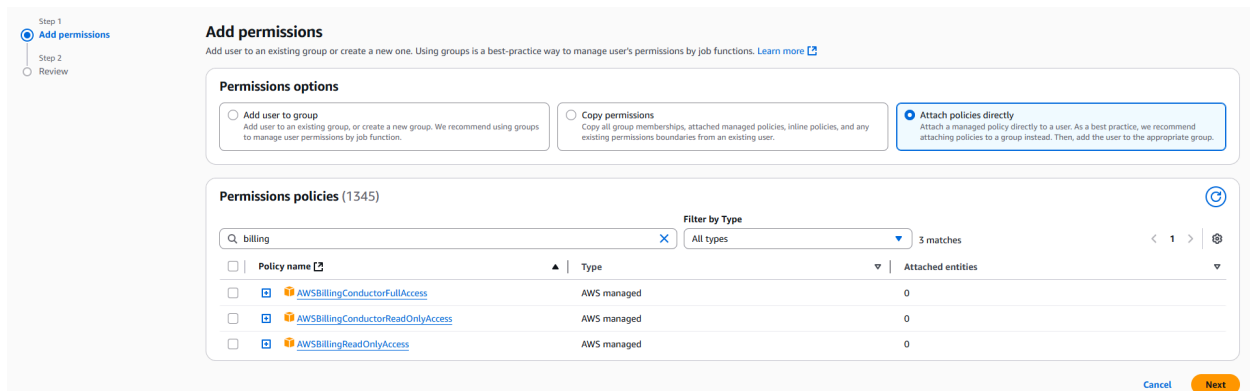
User name	Password	Console sign-in URL
iam-admin-billing-test	CdE98W**	https://awsprimary-acct.signin.aws.amazon.com/console



Billing and Cost Management (Access Denied):

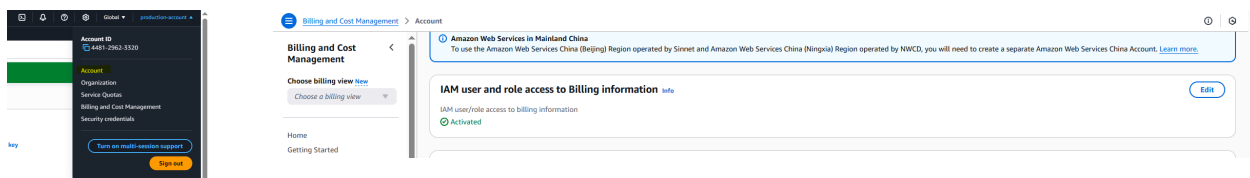


Provided Billing Access:



Billing and Cost Management (Access Enabled):

1. Log in as the **root user** (or a user with root-level access).
2. Go to **Account** → scroll down to “**IAM User and Role Access to Billing Information.**”

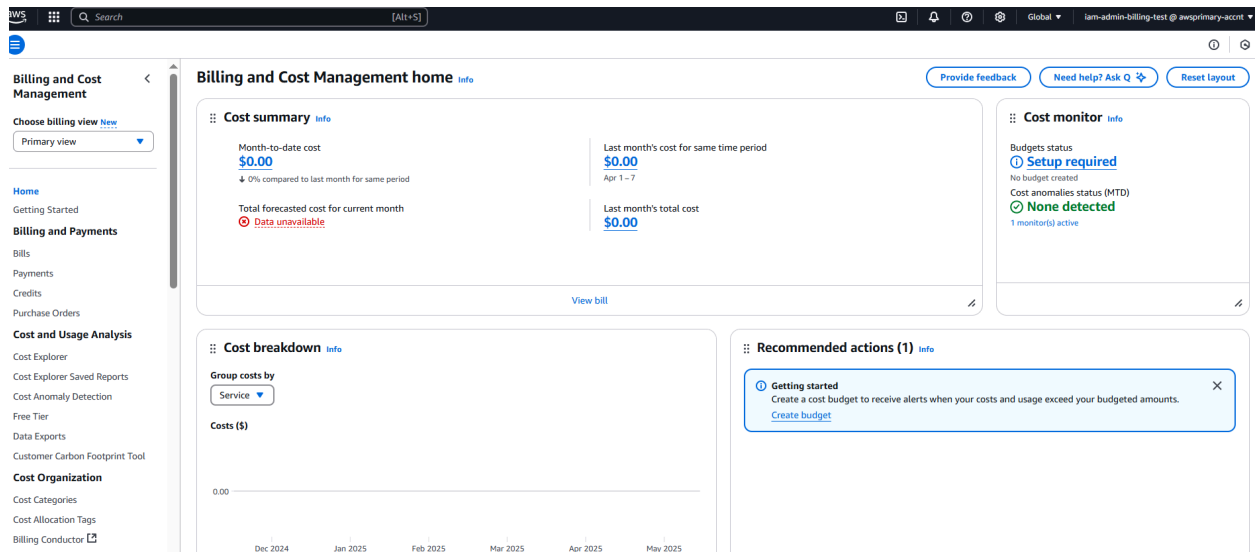


3. Click **Edit**, and check the box:

☒ **"Activate IAM Access to Billing Information".**

4. Click **Update** to save the setting.

Once this is enabled, IAM users with the appropriate permissions (e.g., **AdministratorAccess**) can access the billing dashboard.



Task 2 : create an IAM user, Add "S3FullAccess", login as this user and create a bucket. Now navigate to IAM service and try to create an IAM user and verify.!

Step 1: IAM User Creation

User name	Password	Console sign-in URL
s3-restricted-iamtest	gE[u_Q1**	https://awsprimary-accnt.signin.aws.amazon.com/console

IAM > Users > Create user

Step 1
Specify user details
Step 2
Set permissions
Step 3
Review and create
Step 4
Retrieve password

Review and create

Review your choices. After you create the user, you can view and download the autogenerated password, if enabled.

User details

User name
s5-restricted-iamtest

Console password type
Autogenerated

Require password reset
Yes

Permissions summary

Name

Type

Used as

AmazonS3FullAccess
AWS managed
Permissions policy

Tags - optional

Tags are key-value pairs you can add to AWS resources to help identify, organize, or search for resources. Choose any tags you want to associate with this user.

No tags associated with the resource.

Add new tag

You can add up to 50 more tags.

Cancel Previous Create user

Step 2: Creating an S3 bucket

Amazon S3 > Buckets

Successfully created bucket "iam-s3-bucket-demo"
To upload files and folders, or to configure additional bucket settings, choose [View details](#).

Account snapshot - updated every 24 hours All AWS Regions
Storage lens provides visibility into storage usage and activity trends. Metrics don't include directory buckets. [Learn more](#)

View Storage Lens dashboard

General purpose buckets | Directory buckets

General purpose buckets (1) All AWS Regions

Buckets are containers for data stored in S3.

Find buckets by name

Name	AWS Region	IAM Access Analyzer	Creation date
iam-s3-bucket-demo	US East (Ohio) us-east-2	View analyzer for us-east-2	May 7, 2025, 21:50:16 (UTC-05:00)

Step 3: Create IAM User

IAM > Users

Identity and Access Management (IAM)
Search IAM
Dashboard
Access management
User groups
Roles
Policies
Identity providers
Account settings
Root access management
Access reports
Access Analyzer
External access
Unused access

Users (0)

An IAM user is an identity with long-term credentials that is used to interact with AWS in an account.

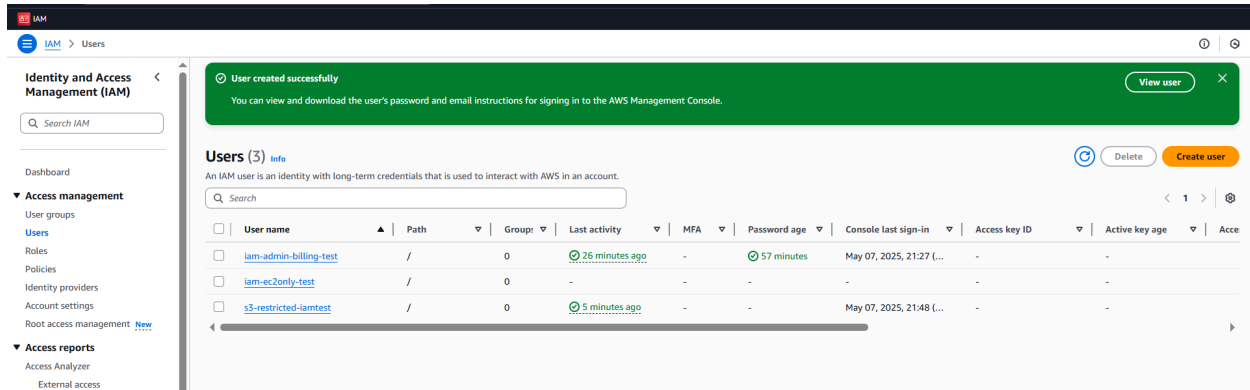
Search

User name	Path	Group	Last activity	MFA	Password age	Console last sign-in	Access key ID	Active key age	Access key last used
Access denied You don't have permission to <code>iam:ListUsers</code>. To request access, copy the following text and send it to your AWS administrator. Learn more about troubleshooting access denied errors. <pre>User: arn:aws:iam:448129623320:user/s3-restricted-iamtest Action: iam:ListUsers On resource(s): arn:aws:iam:448129623320:user/ Context: no identity-based policy allows the action</pre>									

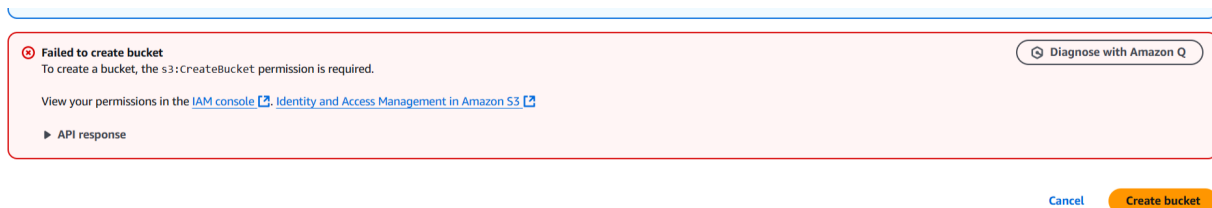
Task 3 : create an IAM user, Add "EC2FullAccess", login as this user and try to create a bucket.

Step 1: Create IAM User

User name	Password	Console sign-in URL
iam-ec2only-test	ndX8@zE_*	https://awsprimary-accnt.signin.aws.amazon.com/console



Step 2: Create S3 Bucket



Task 4: Setup password standard and account alias for your AWS account

IAM → Account Settings under Access Management → Edit the "Password policy"

- Navigate to **IAM** in the AWS Console.
- Under **Access Management**, go to **Account Settings**.
- Click **Edit** next to **Password policy** to configure your organization's password standards (e.g., minimum length, complexity, expiration rules).
- You can also set a custom **Account Alias** here, which replaces the default AWS account ID in the login URL with a more user-friendly name (e.g., `yourcompany.signin.aws.amazon.com/console`).

IAM
Account Settings

Identity and Access Management (IAM)

Search IAM

Dashboard

Access management

User groups
Users
Roles
Policies
Identity providers
Account settings
Root access management

Access reports

Access Analyzer
External access
Unused access
Analyzer settings
Credential report
Organization activity
Service control policies
Resource control policies

Account settings

Password policy

Configure the password requirements for the IAM users.

Edit

Edit password pol

This AWS account uses the following custom password policy:

Password minimum length
8 characters

Password strength

- Require at least one uppercase letter from the Latin alphabet (A-Z)
- Require at least one lowercase letter from the Latin alphabet (a-z)
- Require at least one number
- Require at least one non-alphanumeric character

Other requirements

- Password expires in 60 day(s)
- Allow users to change their own password
- Prevent password reuse from the past 2 changes

Security Token Service (STS)

STS is used to create and provide trusted users with temporary security credentials that can control access to your AWS resources.

Session Tokens from the STS endpoints

AWS recommends using regional STS endpoints to reduce latency. Session tokens from regional STS endpoints are valid in all AWS Regions. If you use regional STS endpoints, no action is required. Session tokens from the global STS endpoint (https://sts.amazonaws.com) are valid only in AWS Regions that are enabled by default. If you intend to enable a new Region for your account, you can use session tokens from regional STS endpoints or activate the global STS endpoint to issue session tokens that are valid in all AWS Regions.

Global endpoint
Valid only in AWS Regions enabled by default | Change

Regional endpoints
Valid in all AWS Regions

Endpoints (10) ...

IAM
Account Settings
Edit password policy

Edit password policy

Password policy

IAM default
Apply default password requirements.

Custom
Apply customized password requirements.

Password minimum length.
Enforce a minimum length of characters.

8
characters

Needs to be between 6 and 128.

Password strength

- Require at least one uppercase letter from the Latin alphabet (A-Z)
- Require at least one lowercase letter from the Latin alphabet (a-z)
- Require at least one number
- Require at least one non-alphanumeric character (!@#\$%^&*()_+-=[\|'`)

Other requirements

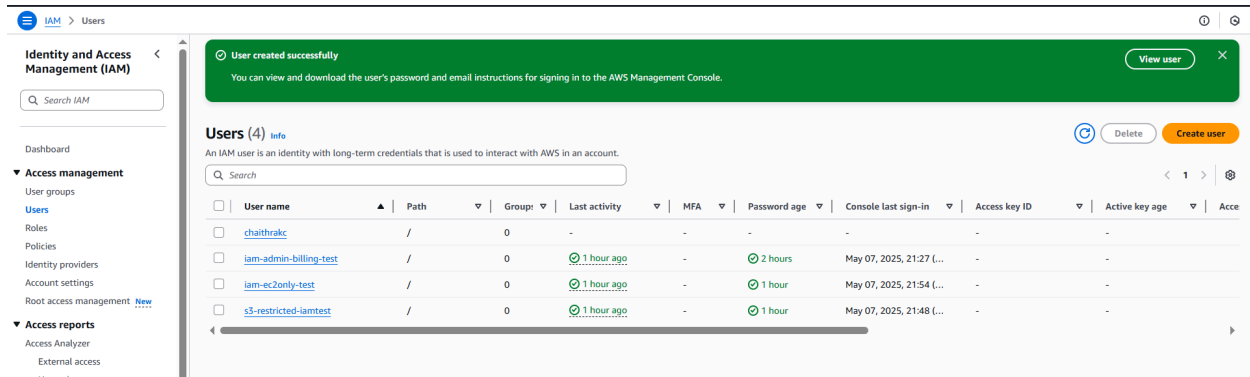
- Turn on password expiration
- Password expiration requires administrator reset
- Allow users to change their own password
- Prevent password reuse

Remember 2 password(s)
Needs to be between 1 and 24.

Cancel
Save changes

IAM User Creation

User name	Password	Console sign-in URL
chaithrakc	xvV!0{Q}*	https://awsprimary-accnt.signin.aws.amazon.com/console



Implicit Allow Mechanism

Access is restricted to only the allocated services. Any attempt to access unassigned services will be denied automatically.

AWS IAM uses an **implicit deny** by default:

- **Users have no permissions unless explicitly granted.**
- Any attempt to access a service or resource that is not explicitly allowed will be **automatically denied**.
- This ensures secure, least-privilege access by default.

Explicit Deny Mechanism

Policy

A policy is a document that defines a **set of permissions** for AWS services, controlling which actions can be performed and on which resources.

When we are creating IAM user, we are associating policy. Policy is nothing but set of permissions on our AWS account. Based on the common use cases, AWS is automatically given the policies. For example:

IAM Password Policies

Creating Customer Managed Policies



AWS managed policies cannot be deleted or modified.

Policy Generator

IAM Policy Simulator

Auditing User Activity